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ONE WORD, MANY APPROACHES

A TRANSATLANTIC LOOK AT COMMERCIAL SPACE PROCUREMENT

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TABLE OF CONTENTS

1 ONE WOR(L)D MANY APPROACHES 3

1.1 What are the objectives of the study? 4

1.2 What is in the scope of the study? 4

2 CONCEPTUAL FRAMEWORK 8

2.1 How to define and address a Commercial Procurement Model? 8

2.2 Which case studies will be analysed? 11

2.3 What are the public policy objectives behind a Commercial Procurement Model? 14

2.4 What methodological framework will be used for the transversal analysis? 16

3 CONCLUSION 18

ACKNOWLEDGMENT & AUTHORS 19

1 One Wor(l)d Many Approaches

"Commercial" is everywhere in the space sector today, becoming shorthand for aspirations of shared abundance, a marker for space programme maturity, and a means of mission assurance through diversity of competition. Encouraged by the expansion of private companies creating innovative space solutions, policymakers increasingly use the term to define their strategy for bolstering economic growth, expanding national capabilities, and improving resilience. This phenomenon is happening across the globe but is especially prevalent in the United States and Europe, two traditional space powers attempting to maintain or reestablish their lead over rising space powers.

Despite important differences in the evolution of their respective space sectors and capabilities, **the United States and Europe both recognise commercial space companies as vital to their economic and military competitiveness**.[1] In the United States, NASA has built sufficient confidence with its commercial cargo and crew programmes to turn to commercial solutions for key components of the Artemis missions to return to the Moon. The United States Space Force, which released its first "Commercial Space Strategy" in 2024, has greatly increased its spending on commercial space services and has overhauled some critical weapons programmes, prioritising available commercial solutions. Similarly, Europe is driving multiple efforts at the national and regional level to both encourage and commercialise space innovation. Examples such as the ESA's ScaleUp, 3CPO, and ELC programmes and the EU's CASSINI provide a variety of resources to help foster and accelerate innovative commercial space companies and entrepreneurs.

Yet across all these announcements, initiatives and new budget lines, **an important question remains – what exactly does "commercial" mean?** The answer may seem obvious, but an examination of commercially-aligned initiatives and missions suggests it is both complex and unsettled. Most people associate "commercial" with products or services they can purchase on the open market, such as a car or laptop, where there are many buyers and sellers. While this is indeed happening today in some areas of space, particularly for well-established markets such as satellite communications services or remote sensing imagery, it hardly describes the whole space ecosystem. Many government space initiatives labelled "commercial" seek to develop capabilities that do not currently exist, such as crewed lunar landers, or for which the government is the only likely customer, such as space-based interceptors. This suggests a much wider definition for "commercial" than assumed and that the term is sometimes used flexibly to support different policy and strategic objectives. In this sense, **"commercial" should not be confused with consumer-driven markets**; it can also apply within government-anchored markets.

[1] For example, the U.S. space sector was long dominated by a small number of large aerospace and defense companies (primes) that designed and built custom solutions for government customers. This legacy has contributed to the challenge of integrating commercial solutions into government missions, which is a necessary demand signal to encourage further funding and innovation. In Europe, government demand and funding has been more limited, while the need for national partnerships and consortia-based solutions to access regional funds complicates solutions. See Sandra Erwin, "Space Force to select multiple companies for new geostationary satellite surveillance network," SpaceNews, July 22, 2025. (Link)

1.1 What are the objectives of the study?

The contemporary interpretation of commercial procurement is deeply related to the 'New Space' paradigm, in which new companies are developing space capabilities, technologies, and systems for both government and commercial customers, **taking on a significant share of the risk in the development of said solutions**. This has brought, among other things, innovative public procurement and support mechanisms to the centre of the debate around the commercialisation of space. In essence, the question of what the government's role in commercial space was reopened and is in search of new answers.

This project aims to provide more clarity to this question of how governments on both sides of the Atlantic are leveraging commercial space. It is a **joint effort between The Aerospace Corporation's Center for Space Policy and Strategy (CSPS) and the European Space Policy Institute (ESPI)**, two global leaders in space policy analysis, that brings together experts from both institutions to examine this question of why and how the United States and Europe are leveraging commercial space.

The project consists of two phases. Phase 1, herein, proposes a framework for characterising and classifying government commercial procurement initiatives and contractual strategies. It proposes three models with different drivers and characteristics: **Commercial-Lite, Commercial-Led, and Purely Commercial**; and applies them to a set of example efforts in the United States and Europe.

Programme Characteristics	Commercial Procurement Model		
	Commercial-Lite Approach	Commercial-Led Approach	Purely Commercial Approach

Figure 1: Summary of the Commercial Procurement Model

The forthcoming Phase 2 of the project will perform a full cross-case comparison of the regional examples and make recommendations on how to align and structure types of future commercial programmes to government policy drivers.

1.2 What is in the scope of the study?

The participation of private companies in government space programmes is not new – from the very beginning of the Space Age, the United States and Europe have relied on private sector capabilities. However, the more recent developments where governments are not only relying on private contractors to perform work but **shifting a greater share of choices and decisions to the private sector represent a significant change in philosophy**. With a shift in Western economies throughout the late 1970s, space increasingly became understood as a domain that can not only provide complementary value to activities on Earth, but itself drive economic growth and (through the advent of global markets)

empower industrial competitiveness. Several approaches have been explored in this study to emphasise limited government involvement, cost sharing, ownership of intellectual property by the commercial partner, top-line requirements, pay-for-performance milestones (including company ownership in devising the milestones and associated payment) and lower costs.[2]

In the United States, national space policy has for decades supported the development of commercial space capabilities and encouraged government entities to make use of them. These include the National Space Policy signed by President Reagan in 1988 that directed government agencies to "purchase commercially available space goods and services to the fullest extent wherever possible."[3] However, implementation of these policies in the following decades, particularly for major space programmes, was mixed at best. It was not until the success of NASA's Commercial Orbital Transportation Services (COTS) and the follow-on Commercial Resupply Services (CRS) in the 2000s and 2010s that there was a significant shift in the relationship between U.S. government agencies and the private sector for civil space activities. On the defence side, the National Geospatial Intelligence Agency's 2015 "Commercial GEOINT Strategy" was a strong public signal of a similar shift.[4] These new arrangements were very different from the traditional, heavily regulated contracts, where the opposite was often the case, such as the government providing detailed programme design requirements.[5]

In Europe, the broader question of commercialisation is also not as recent as it is commonly understood. In January 1980, France submitted a declaration to the ESA Council to transfer the "commercial responsibility of manufacturing, marketing and launch operations of Ariane" from ESA to a newly created company called Arianespace. Looking towards downstream markets, in 1981, a programme for the first ESA Remote Sensing Satellite (ERS-1) was agreed upon whereby it (i.a.) hoped to "*benefit a number of commercial applications (offshore activities, ship routing, ice surveillance, aid to fisheries etc.).*"[6] Simultaneously, as the French commercial satellite imagery company Spot Image was being conceived, supported by a public programme, hopes emerged of a future market that could uptake between 60.000 and 100.000 images per year at an estimated cost of $1.000 to $2.000 per image, with primary users coming from (i.a.) the oil & gas, mining, and agricultural industries.[7] Beyond programmatic developments, the ESA Office of Space Commercialisation was also created in 1986 to identify ways of preparing for the commercialisation of ESA programmes and services.[8]

There are different objectives in devising this new government-industry relationship, but at

[2] NewSpace: The "Emerging" Commercial Space Industry, NASA, 2014, (link)
[3] "Fact Sheet: Presidential Directive on National Space Policy", The White House, February 11, 1988. (Link)
[4] "NGA rolls out commercial GEOINT strategy". Amber Corrin, CS4ISRnet, Oct 29, 2015 (Link)
[5] Developing Cislunar Space Using the COTS Model, White Paper by Bruce Pittman & Dr. Daniel J. Rasky
[6] *Europe's Place in Space*, by European Space Agency (Paris, 1981) at 27.
[7] Le Monde, "60 000 à 100 000 images à vendre", *Le Monde.fr* (11 November 1981), (Link).
[8] European Space Agency, *The ESA Programmes (BR-114)* (European Space Agency, 1995).

its crux sits the development of a robust, innovative and competitive commercial space sector, built on an **industry-led and government-reinforced strategy**, where the latter refrains from competing or discouraging commercial space activities.[9]

Building on these notions, as a starting point, we define commercial space procurement as the public acquisition of space-related products and services from private sector enterprises that leverage **market-based dynamics** and a **degree of cost and risk sharing**, rather than bespoke, agency-prescribed development to deliver products for government use. In some cases, but not all, there is also an explicit policy goal of using public demand to enable firms to provide their services and products to a broader market of non-governmental customers.[10] There are limitations to a single unidimensional definition of what commercial procurement is, but **this starting point enables us to delineate what falls within and outside our scope of study.**

Procurement Programme Characteristics		Bespoke Development Model	Commercial Procurement Model
Owner	Ownership over the product/service procured	Government	Industry
Contract Type	Type of contract	Cost-Reimbursement[11]	Fixed Price[12]
Contract Management	Industry contractual relationship with the government	Prime Contractor	Product or service provider/partnership
Customer(s)	Type of customers that the product/service aims to gather	Government	Government and non-government
Funding for Capability Demonstration	Type of support for the development of the capabilities	Government procures capability	Government provides investment via milestone payments
Government's Role in Capability Development	Level of ownership by the industry in capability development	Government defines "what" and "how"	Government defines "what", industry defines "how"
Cost & Risk Structure	Level of cost & risk sharing	Government incurs total cost and most of the risk	Government and industry share cost, and risk falls mostly with industry

Figure 2: Bespoke Development Model (out of the scope of this paper) and Commercial Procurement Model (Source: NASA, ESPI)

[9] That is, to develop the "commercial (space) sector" which was defined in the "National Space Policy of the United States of America," June 28, 2010, as "services, or activities provided by private sector enterprises that bear a reasonable portion of the investment risk and responsibility for the activity, operate in accordance with typical market-based incentives for controlling cost and optimizing return on investment, and have the legal capacity to offer these goods or services to existing or potential nongovernmental customers commercial." See also, The Rise of the Commercial Space Industry: Early Space Age to the Present (link).

[10] Commercial Orbital Transportation Services: A New Era in Spaceflight—Lyndon B. Johnson Space Centre Staff; ESPI, Commercial Space Exploration Potential Contributions of Private Actors to Space Exploration Programmes

[11] Cost Plus: A contracting mechanism that allows an entity to develop a technology and be reimbursed post-delivery. This mechanism is subject to adjustment based on development costs (Link)

[12] Fixed Price: the contracted entity is paid an amount agreed upon before production (Link)

There is an important nuance for the risk structure under the commercial procurement approach. While industry does share more of the contract risk, the government may still bear the consequences of not having a critical capability should the contract fail to deliver. For example, if the contract is to deliver a critical component for a programme, failure could jeopardise the entire programme's success and resolving it may require additional time that the government does not have.

To better understand what, in practice, can be considered a Commercial Procurement Model, and thus what is in scope of this study, it is useful to start by identifying what clearly falls outside this scope. On the European side, the Automated Transfer Vehicle (ATV) developed by ESA to transfer cargo to and from the International Space Station (ISS) represents a typical case of a Bespoke Development Model (full public ownership, cost–reimbursable contracting, and top–down management) and therefore lies beyond the analytical scope of this research. The programme was owned and funded by ESA, fulfilling Europe's contribution to the ISS partnership. The contract structure followed a cost-reimbursable model, with ESA covering all development and production expenses and retaining full financial and performance responsibility. As for the contractual relationship, the industrial partner, Aerospatiale (now Airbus D&S), acted as prime contractor executing ESA technical directions, under a scheme where decision–making remained with the Agency. The customer base was governmental, and funding came entirely from public budgets, with no risk-sharing mechanisms. In terms of capability development, ESA defined both the objectives and the technical requirements, leaving marginal scope for industrial discretion or innovation. Finally, the **cost & risk structure** was wholly public, confirming the absence of commercial elements in the programme design and execution.[13]

On the U.S. side, an example of the Bespoke Development Model is the Global Positioning System (GPS) III program, which began in the early 2000s to both sustain the operational GPS constellation and deliver new capabilities. Following competition, the U.S. Air Force awarded the contract for GPS III to Lockheed Martin, a traditional contractor that had built prior-generation GPS satellites. The Air Force used cost reimbursement contracts to fund GPS III research and development (R&D), as well as most procurement activities. Because GPS signals are available for commercial use at no charge, the government functions as the sole customer for GPS III. Similarly, the government defined the program's objectives and technical requirements, and the cost and risk structure was wholly public.

[13] European Space Agency, *Automated Transfer Vehicle* (Link)

2 CONCEPTUAL FRAMEWORK

While the aforementioned definition of commercial procurement is sufficient to separate it from more established bespoke approaches, it does not clearly identify which mechanisms qualify as commercial procurement or capture the nuances of European and U.S. ecosystems. To address this, we further expand the initial definition into a spectrum-based framework that reflects the distinctive features of each ecosystem and recognises that individual programmes may combine multiple models. For instance, a programme may align mainly with the Commercial Procurement Model while incorporating elements of the Bespoke Development Model; however, if its dominant characteristics reflect the latter, it is not included in this analysis.

2.1 How to define and address a Commercial Procurement Model?

Figure 2 demonstrates how we further divide the Commercial Procurement Model into three main categories or approaches with different programme characteristics. The reasoning behind each approach (**Commercial-Lite, Commercial-Led, and Purely Commercial**) was informed by relevant documents (e.g. the United States Federal Acquisition Regulation, ESA Contracting Regulation 2025) and by identifying gaps in the Bespoke-Commercial paradigm across European and United States procurement programmes. **Procurement programmes often reflect more than one approach**, and are most likely to be multi-dimensional, leaning more "purely commercial" on some characteristics, while being more in line with "commercial-Lite" on other characteristics. Nevertheless, overall, the more a programme characteristic leans into **supporting market-based dynamics**, attributing ownership of the product and/or service design, build, financing and operation, to the private enterprise, and thus **transferring associated risks away from the government**, the more purely commercial the programme will be. We note that under these definitions, there are very few current space procurement efforts that would fall under the purely commercial approach.

Programme Characteristics	Commercial Procurement Model		
	Commercial-Lite Approach	**Commercial-Led Approach**	**Purely Commercial Approach**
Ownership	Industry with limited control	Government-enabled ownership	Full industry ownership
Contract Type	Fixed-Price with Cost Reimbursable Elements	Fixed Price with adjustable price	(Firm-)Fixed Price
Contract Management	Multiple Primes; Traditional PPP	Innovation-driven PPP or service	Service or Product Provider

Customer(s)	Government-locked	Government and Non-government	Non-government and Government
Funding for Capability Demonstration	Capability development is baked into the programme	Government can have separate but connected programmes to help develop the capability / buy COTS adapted to needs	None (pure Commercial off-the-shelf; COTS)
Government´s Role in Capability Development	Government defines "what", government and industry define "how"	Government defines "what", industry defines "how"	Government buys what is available
Cost & Risk Structure	Government bears most cost & risk	Government and industry share cost & risk	Industry bears most cost & risk

Figure 3: The Spectrum of the Commercial Procurement Model

Although some of the programme characteristics are self-explanatory, there are still some important considerations that are explored below.

Ownership

In the Commercial-Lite Approach, industry builds and/or operates government infrastructure or delivers exclusive services under government direction. This role enables the company to scale capabilities and to later offer its own derivative or related products to the broader market. In the Commercial-Led Approach, the company owns its product or service from the outset, while the government provides specific enablers that reduce development risk. Typical enablers include industry access to infrastructure, facilities, and data, targeted technology transfer, spectrum access, and milestone-based funding. In a Purely Commercial Approach, the industry has full ownership over its solutions/products without government intervention.

Contract Type

The contract type is a good indicator to assess the level of risk transfer to the private enterprise. The Commercial-Lite Approach combines a fixed price for part of the work with the reimbursement of specific, allowable costs for other elements of the work. The Commercial-Led Approach relies on fixed-price contracts with adjustable prices, meaning the private firm bears a substantial part of the risk but is still shielded from external pressures because it allows for adjustments based on predefined criteria, such as fluctuations in labour or material costs. In the Purely Commercial Approach, the firm bears all the risk because the price of the contract cannot be altered or revised based on the actual costs incurred by the company in fulfilling its business obligations.[14]

[14] ECSS, Space project management, ECSS-M-ST-60C, 2008.

Contract Management

Contract management hinges on two main concepts: *i)* type and quantity of firms contracted (e.g., multiple primes), and *ii)* what framework governs the government/industry relationship.

The degree of openness in the selection process largely determines the structure of the supplier base and signals the government's intended relationship with industry. When the priority is short-term value for money or the delivery of a bespoke project, the government will typically select a single prime contractor. A sole integrator can capture economies of scale more effectively than a fragmented set of providers and is more likely to field mature, ready-to-apply technologies. By contrast, if the aim is to stimulate market development in the space sector, the process should be open and competitive, replicating market dynamics by selecting multiple firms and/or ensuring competition in future contract awards. In that setting, the government may play a broader role – as investor, technical adviser, and partner – helping develop solutions and bring them to market.[15]

These choices also shape the Public-Private Partnership (PPP) design, when there is one. Two considerations are central here. First, is the level of risk transfer to the private entity, which depends, among other factors, on who ultimately owns the asset and whether payment guarantees (such as minimum revenue guarantees or availability payments) are in place.[16] Second, is the rationale of the PPP. Traditional PPPs are organised around value for money, while innovation-oriented PPPs prioritise solving complex problems and commercialising outcomes by pooling public and private capabilities.[17]

In the Commercial-Lite Approach, the government works with multiple prime contractors, which can be done under a traditional PPP. In the Commercial-Led Approach, the government focuses on supporting innovation through service-driven contracts or innovation-focused PPPs. In the Purely Commercial Approach, the contract management is centred on the private enterprise as a provider of the service or product, more akin to a pure commercial transaction where the government buys the product or service as needed.

Customers

In the Commercial-Lite Approach, private enterprises openly compete to provide a solution that will be exclusive to the government, although some market dynamics are replicated within the procurement process. In the Commercial-Led Approach, companies first deliver a solution to the government that is then expected to serve non-government customers. The Purely Commercial Approach would be epitomised by the government procuring a solution that is already available to non-

[15] Commercial Orbital Transportation Services: A New Era in Spaceflight—Lyndon B. Johnson Space Centre Staff.
[16] See World bank, Identifying Risks (Link); PPP Contract Types and Terminology (link); Guarantee Mechanisms in PPPs (Link)
[17] OECD, "Recommendation of the Council on Principles for Public Governmance of Public-Private Partnerships; OECD, "Public-Private Partnerships for Agricultural Innovation"

government customers. It should be noted that in the case of space, the public good nature of many products and services means that governments may continue to be a significant portion of the customer base, even under the Purely Commercial Approach.

Funding for Capability Demonstration

This programme characteristic focuses on the type of support for the development of the capabilities being procured. In the Commercial-Lite approach**,** the government funds the R&D necessary to provide the capability as part of the procurement process. In the Commercial-Led Approach, the government asks for a modified version of a solution meant for non-government customers or takes advantage of synergies with other government programmes that helped the R&D process. In either case, the R&D required to develop the company's solution was not made specifically for the government procurement programme in question. In a Purely Commercial Approach, the government buys the solution commercial-off-the-shelf, contributing to the commoditisation of the company's offering.

Government's Role in Capability Development

In the Commercial-Lite Approach, the government establishes only top-level but extensive requirements, limiting the company's degree of ownership over how the solution is developed and delivered. In the Commercial-Led Approach, the government specifies capabilities it requires and aims to minimise top-level requirements, effectively allowing companies to determine the best way to achieve those capabilities. In the Purely Commercial approach, the government simply procures the solutions already available on the market that are best aligned with its needs. Governments also have long had programmes to fund research and early-stage capability development by private sector entities separate from procurement.

Cost & Risk Structure

Regarding Cost & Risk Structure**,** in a Commercial-Lite Approach, the government pays most of the development costs and takes on most of the risk, while industry contributes less and follows government direction. In a Commercial-Led Approach, costs and risks are shared more evenly, and the government helps de-risk specific phases by providing facilities, data, testing infrastructure, or milestone payments for capability development. In a Purely Commercial model, industry funds the development of its solutions and carries nearly all the risk, while the government is one of many customers with only typical buyer risk.

2.2 Which case studies will be analysed?

This paper adopts a case study methodology designed to identify and assess procurement patterns that are intended to catalyse market dynamics in both the United States and Europe. The methodology follows a functional logic, focusing on how procurement mechanisms operate across institutional contexts and how they contribute to several policy objectives. The selection of case studies represents a variety of parameters such as

geographical and functional types (various government agencies), use (civilian, dual-use, and military), applications/domains (i.e. remote sensing, satellite communications, space situational awareness, and space transportation) and procurement models present in Europe and the United States. Within this framework, six European programmes are presented below.

Case Studies	Promoting Entity	Strategic Objectives	Market Interface[18]	Contextual Information[19]
LEO Cargo Return	ESA	Development of commercial cargo transport to/from LEO	Structured – industrial co-development with potential for future market use	+-€75M (phase 1) CIVIL ONGOING
IRIDE	MS	National constellation for environment, security, emergency services	Structured – competitive industrial procurement with open downstream market for data use in turn for operational cost bearing	+-€1.1 B DUAL USE ONGOING
IRIS²	EU	Secure, autonomous satellite connectivity; commercialisation of operator services	Structured – competitive PPP (concession contract) with partial market exposure and potential for service commercialisation	+-€10.6B DUAL USE ONGOING
SKYNET 6	MS	Military SATCOM ensuring resilience and strategic effects	Structured – mixed procurement scheme introducing competition and partial risk-sharing within a government-funded programme	+-€5.9B DEFENCE ONGOING
EMISSARY	EU	Development of European military space situational awareness	Structured – cooperative public procurement with limited industrial competition and indirect market exposure through dual-use potential	+-€157 M DEFENCE ONGOING
EECCM	EU	Contributing EO missions to Copernicus, strengthening EU data provision	Open – private operators sell data services directly to public	+-€65M per year between 2021 and 2027 CIVIL ONGOING

Figure 4: European Case Studies Overview

As discussed earlier, the United States has prioritised developing a commercial space industry since at least the 1980s and leveraging commercial procurement since the 1990s. These efforts were central to national space policies issued by various Presidential administrations, though implementation was largely left to the individual agencies with Congressional oversight and direction. In general, the United States defines “commercial”

[18] By the term market interface we refer to the degree and form of private-sector participation, including innovation, competition, and opportunities for commercialisation.
[19] Context Information includes approximate budget, functional orientation and timeline.

based on *who* provides the product or service, as opposed to *what* the item is or *how* it applies to the commercial market. Six U.S. programmes are presented below.

Case Studies	Promoting Entity	Strategic Objectives	Market Interface[20]	Contextual Information[21]
Commercial Data Program (CDP)	NOAA	Acquiring commercial, space-based radio occultation data for use in NOAA's products and services	Open – vendors meeting instrument and data requirements compete for individual task orders	+-$124M (combined buys spanning 2020-2028)[22]
Commercial Crew Program (CCP)	NASA	Partnership with private industry to provide transportation to the ISS.	Structured – competitive PPP with shared risk and demand	**Per Seat**: $55M (SpaceX); $90M (Boeing).[23] **Lifetime total in 2019**: $4.3B (Boeing); $2.5B (SpaceX).[24] **Total awarded for CCP**: $8.5B.[25]
Next-generation Exploration Extravehicular Mobility Unit (xEMU)	NASA	Provide spacesuits for the ISS and Artemis missions	Structured – competitive industrial procurement of task orders	$3.5B across program for both missions; $97.2M for ISS suit, $5M to advance capabilities.[26]
Transport Layer, Tranche 1	DOD	Data transport system supporting several military missions	Structured – competitive industrial procurement with opportunities for related commercial applications	+-$3B Fielding began September 2025
Protected Tactical SATCOM – Global (PTS-G)	DOD	Space and ground SATCOM system with proliferated constellation in Geosynchronous orbit for moderate degree of assured access	Structured – competitive industrial procurement—maximising COTS solutions—with opportunities for related commercial applications	+-$4B ceiling

[20] By the term market interface, we refer to the degree and form of private-sector participation, including innovation, competition, and opportunities for commercialisation.
[21] Context Information includes approximate budget, functional orientation and timeline.
[22] E.g. of data buy is NOAA, *NOAA Awards First Commercial RO Contracts Supporting Operational Weather Forecasting* (Link)
[23] NASA, *NASA's Management of crew transportation to the International Space Station* (Link) at 6
[24] NASA, *NASA's Management of crew transportation to the International Space Station* (Link) at 4
[25] NASA, *NASA's Management of crew transportation to the International Space Station* (Link) at 1 (NASA recommendations for improvements)
[26] Axiom, *Axiom Space wins NASA Contract to Build Next Generation Astronaut Spacesuits* (Link); NASA, *NASA Collins xEVAS Update* (Link)

Commercial Proliferated LEO Services	DOD	Programmatic vehicle for access to a range of commercial services from LEO-based satellites	Open – private operators become approved vendors and compete for individual task orders	+-$13B ceiling (over 5 years)

Figure 5: United States Case Studies Overview

2.3 What are the public policy objectives behind a Commercial Procurement Model?

Commercial procurement plays a crucial role in aligning government and industry capabilities with national interests. Commercial procurement is not only about procuring goods and services but also about ensuring that investments in space deliver long-term national benefits. The policy objectives behind such contracts set the guiding principles that influence how resources are allocated, risks are managed, and strategic outcomes are achieved.

Macro Policy Objectives	Micro Policy Objectives	Definition
Innovation		Stimulates the creation and adoption of new technologies, processes, and services by leveraging the agility of commercial actors.
Cost-Effectiveness		Ensures better value for money by using competitive procurement, leveraging economies of scale, and transferring risk to industry.
Speed		Accelerates development and deployment timelines compared to government-led programmes, meeting urgent capability needs faster.
Lowering Programme Risk		Distributes technical, financial, and operational risks across multiple suppliers, reducing dependency on a single source.
Market & Competitiveness	**Increased Competition**	Encourages multiple companies to compete within the domestic market, driving down costs and improving quality and innovation.
	Commercialisation of space	Promotes private sector-led space activities with the goal of enabling a sustainable commercial ecosystem that can exist independently from government support.
	Scaling	Supports companies in growing beyond pilot projects to large-scale operations, enabling them to meet domestic and international demand.
	Economic development	Uses commercial contracting to stimulate jobs, private investment, and wider growth in the space and adjacent sectors.
	Capturing Global Markets	Positions domestic companies to export technologies and services internationally, strengthening global market share.

Sovereignty & Leadership	**Diversify suppliers**	Reduces reliance on a limited set of providers, strengthening resilience against supply chain disruptions.
	Development of specific capabilities	Focuses investment on key technological areas (e.g., launch, Earth observation, secure communications) to ensure sovereign access.
	Strengthening Champions	Builds and supports nationally/regionally significant companies with a higher degree of competitiveness, in turn, better capability to compete globally and safeguard strategic public interests.
	Industrial base sustainment	Maintains a robust, long-term space industrial base that can support national security, economic, and scientific priorities.

Figure 6: Overview of Selected Public Policy Objectives

It is important to note that these public policy objectives can be either explicit or implicit. **Explicit/outspoken policy objectives** are those openly articulated in official documents such as national space policies, defence and industrial strategies, regulatory frameworks, and high-level public statements by government leaders. **Implicit/indirect policy objectives** may not be formally stated but can be **subjectively identified through transversal policy analysis** that looks across different sectors. These underlying objectives often reflect long-term strategic considerations, shaping geopolitical influence or subtly steering markets toward desired technological pathways. Taken together, these explicit and implicit objectives provide the underlying rationale for a commercial-driven contracting approach in space. They highlight how governments seek to balance immediate operational needs with ambitions for autonomy, leadership and competitiveness in an increasingly contested and economically vital domain.

A further question is whether such policy objectives are intended to **foster existing capabilities** or to **develop new capabilities** within a certain geographical or industrial context. 'Existing capabilities' often reinforce and scale up areas of established national strength, while 'new capabilities' may target emerging or less mature sectors where governments seek to stimulate novel industrial ecosystems.

This distinction is important because it shapes how contracting mechanisms are designed — whether to consolidate market leaders, sustain industrial bases, or to seed and nurture new entrants and breakthrough technologies. For the scope of the paper, they are defined as follows:

Existing Capabilities	Industrial, technological, and operational competencies are already established within a country or region when a commercial procurement model is introduced. They rely on an industrial base with proven processes, supply chains, and skilled expertise. Policy objectives targeting existing capabilities aim to reinforce, scale, or sustain these strengths to preserve autonomy, ensure competitiveness, and secure continuity of supply in strategic sectors. This may

New capabilities	involve upgrading legacy systems, supporting market leaders, or preventing industrial decline. Industrial, technological, or operational competencies that are not yet fully developed or embedded in the local ecosystem but are considered strategically important by policymakers. They may include emerging frontiers of the space economy, disruptive technologies, or areas where there has been prior dependency on third parties. Policy objectives targeting new capabilities are generally aimed at stimulating innovation, incentivising private investment, and lowering market-entry barriers for new actors.

Figure 7: Definition of existing capabilities versus new capabilities

2.4 What methodological framework will be used for the transversal analysis?

Phase 2 of the study will perform a case study analysis across the example commercial procurement efforts and develop a cross-case comparison within and across both Europe and the U.S. to highlight similarities or differences in approaches. This transversal, or cross-case analysis, developed in Phase 2 will explore the interconnections between existing capabilities, the different approaches to commercial procurement, and the policy objectives driving them. It will capture how industrial capacity, strategic priorities, and commercial contracting approaches interact. Policy objectives will also be examined to determine whether they target existing capabilities, foster new ones, or both.

As an early-stage example, the **European Copernicus Contributing Missions** (ECCM) largely follows the Purely Commercial Approach. Satellites are privately (or at a minimum separately) financed, developed, and operated by industry, and companies act as service providers to Copernicus. Purchases are made via ESA's Dynamic Purchasing System using fixed-price call-offs for Earth-Observation data. The programme is not directly linked to government-funded capability development, giving industry strong autonomy to design systems and solutions and shoulder most of the costs and risks. That said, ECCM also shows elements of a Commercial-Led Approach. While ESA does not establish the requirements for the systems ex-ante, it is also not selecting what to buy directly from the market. The providers need to join the ECCM programme, and for that, they need to comply with certain standards. These lead to so-called "restricted competitions". Therefore, it can still input some directionality into the 'what' when firms are designing their solutions. Overall, ECCM is best described as predominantly Purely Commercial, with minor deviations toward a Commercial-Led Approach. This provides a valuable example of how the spectrum of methodology captures programme nuances.

Among the United States case studies, an example of Commercial-Lite procurements for civil space purposes is the NASA **xEMU**, or spacesuit, program. Spacesuit design and requirements differ depending on their purpose, most notably if they are to be worn in a

pressurised environment or require pressurisation of their own. For example, the Extravehicular Mobility Units (EMUs) worn during spacewalks outside the ISS are different from Intravehicular Activity (IVA) suits meant to be worn inside a pressurised craft. Most recently, NASA awarded two companies' contracts for spacesuits to support the ISS and Artemis missions, called xEVAS (Exploration Extravehicular Activity Services) through milestone-based task order contract awards.

The application of this framework to all of the European and United States cases in Phase 2 will serve to compare approaches, identify differences, and derive best practices. Using a matrix mapping across the three approaches, the analysis will provide a nuanced understanding of how a commercial procurement model translates policy objectives into practical industrial and technological outcomes in the space sector, with the ultimate goal to highlight effective procurement approaches, consistent policy objectives, gaps, and opportunities to inform future strategies.

3 CONCLUSION

The analysis presented in this paper demonstrates the breadth and complexity of how both the United States and Europe currently use the term "commercial" to apply to space activities. While both currently put significant emphasis, particularly in public statements, about "going commercial", in reality, that label is being applied to a wide range of programme structures and policy goals that at first glance seem somewhat disconnected from each other.

To provide clarity, we define Commercial Procurement as being distinct from traditional Bespoke Procurement, as characterised by the owner of the capability, the contract type, contract management model, customer(s), funding, and the government's role in the development. Furthermore, we propose three different approaches within Commercial Procurement – Commercial-Lite, Commercial-Led, and Purely Commercial- that are defined by further variation of their characteristics. Finally, we identify multiple macro and micro public policy objectives that are established by governments (either explicitly or implicitly) to ensure that investments in space deliver long-term benefits and achieve strategic outcomes.

This paper also identifies multiple examples of civil or defence space programmes in both the United States and Europe that utilise one of the three Commercial Procurement approaches. For Phase 2 of this study, we plan to conduct a case study analysis to further examine these examples and how they map across public policy objectives in order to more fully understand how commercial procurement models translate policy objectives into practical industrial and technological outcomes. In the process, we will identify both similarities and differences between the United States' and European approaches and any recommended practices.

In preparation for the second phase, we ask the **community of stakeholders for feedback on our proposed definitions, conceptual framework, and selected case studies**. We hope that this feedback will help us refine our work to ensure that it accurately reflects the most important aspects of this critical topic and can help bring more clarity to future discussions of how best to leverage commercial in the future.

If you are interested in contributing, please send us an email at info@espi.eu and policy@aero.org.

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